

Bitwise linear trail through **WARP** found by SAT

CiVerLy

July 21, 2026

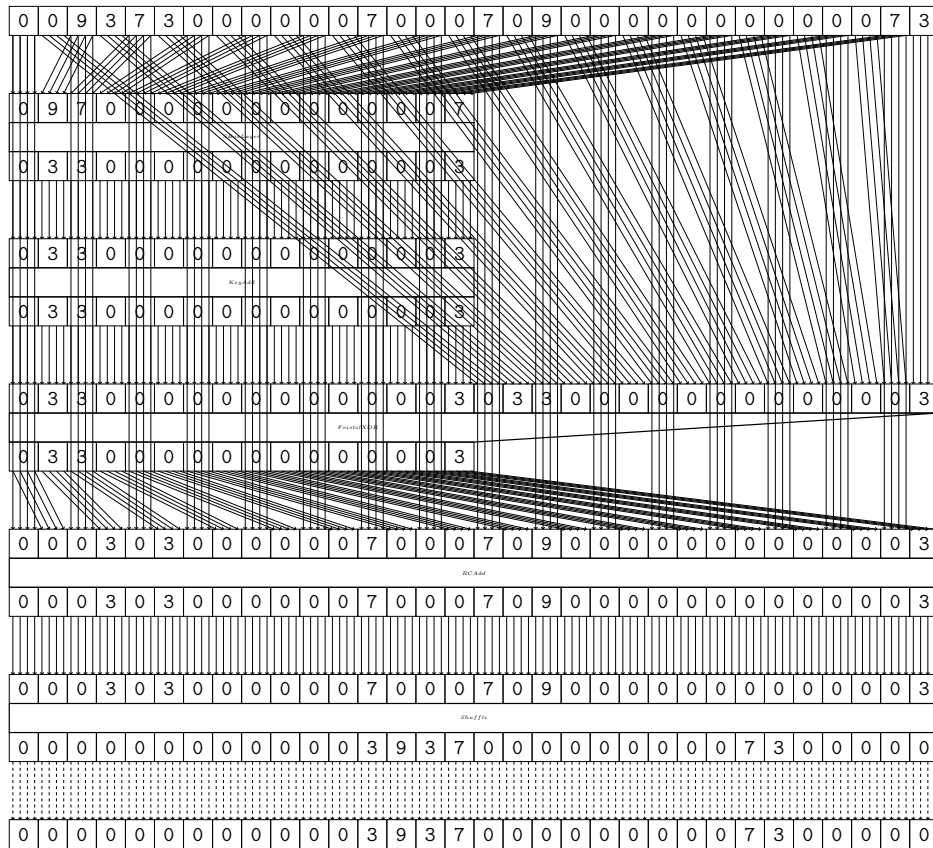
1 WARP

Maximal linear correlation: 2^{-14}

0	0	9	3	7	3	0	0	0	0	0	0	7	0	0	0	7	0	9	0	0	0	0	0	0	0	0	0	7	3		
0	0	9	3	7	3	0	0	0	0	0	0	7	0	0	0	7	0	9	0	0	0	0	0	0	0	0	0	7	3		
unpermut																															
0	0	0	0	0	0	0	0	0	0	0	0	3	9	3	7	0	0	0	0	0	0	0	0	0	0	7	3	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	3	9	3	7	0	0	0	0	0	0	0	0	0	0	7	3	0	0	0	0
unpermut																															
0	0	0	0	9	0	0	0	0	0	7	3	0	0	0	0	0	7	3	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	9	0	0	0	0	0	7	3	0	0	0	0	0	7	3	0	0	0	0	0	0	0	0	0	0	0	0	0
unpermut																															
3	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	
3	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	
unpermut																															
0	0	0	0	0	0	9	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	9	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
unpermut																															
0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
unpermut																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0
unpermut																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0
unpermut																															
0	0	0	0	0	0	0	0	0	0	0	0	3	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0
0	0	0	0	0	0	0	0	0	0	0	0	3	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0
unpermut																															
0	0	0	0	3	0	0	9	0	0	3	0	0	0	0	0	0	0	0	0	0	0	7	0	7	0	0	0	0	0	0	0
unpermut																															
0	0	0	0	3	0	0	9	0	0	3	0	0	0	0	0	0	0	0	0	0	7	0	7	0	0	0	0	0	0	0	0

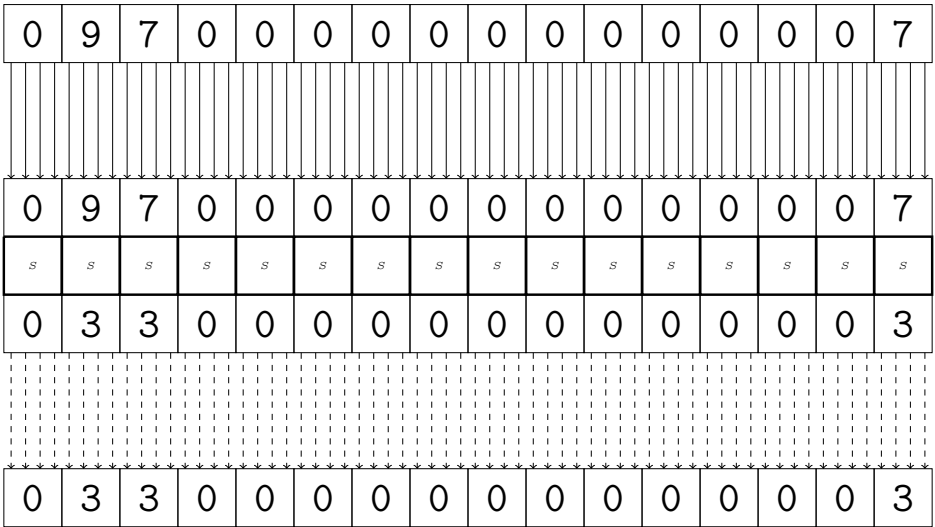
2 warp_round

Maximal linear correlation: $2^{-3.0}$

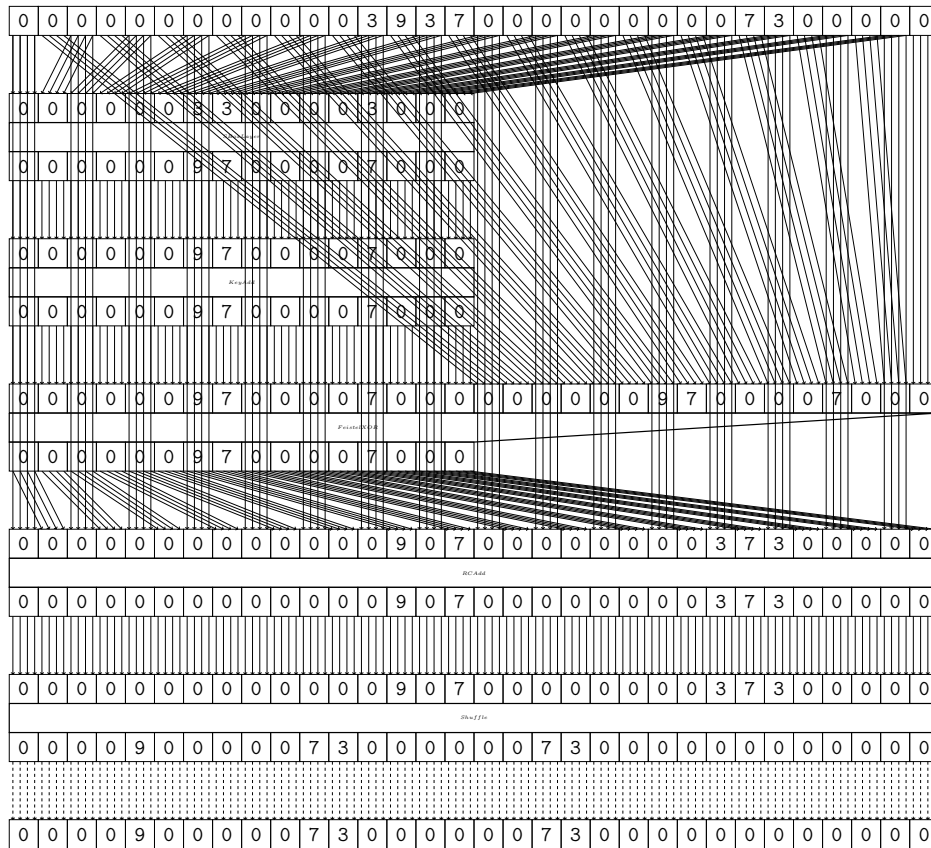


3 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

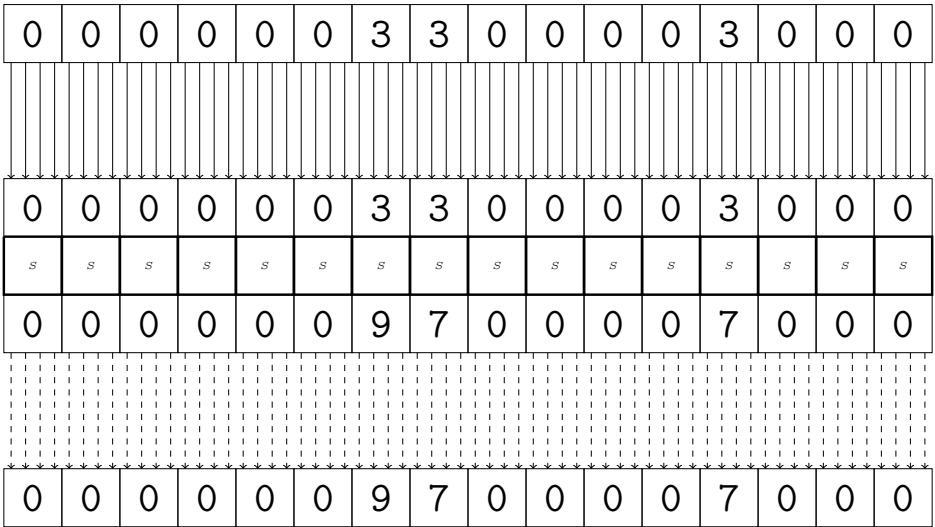


4 warp_round

Maximal linear correlation: $2^{-3.0}$ 

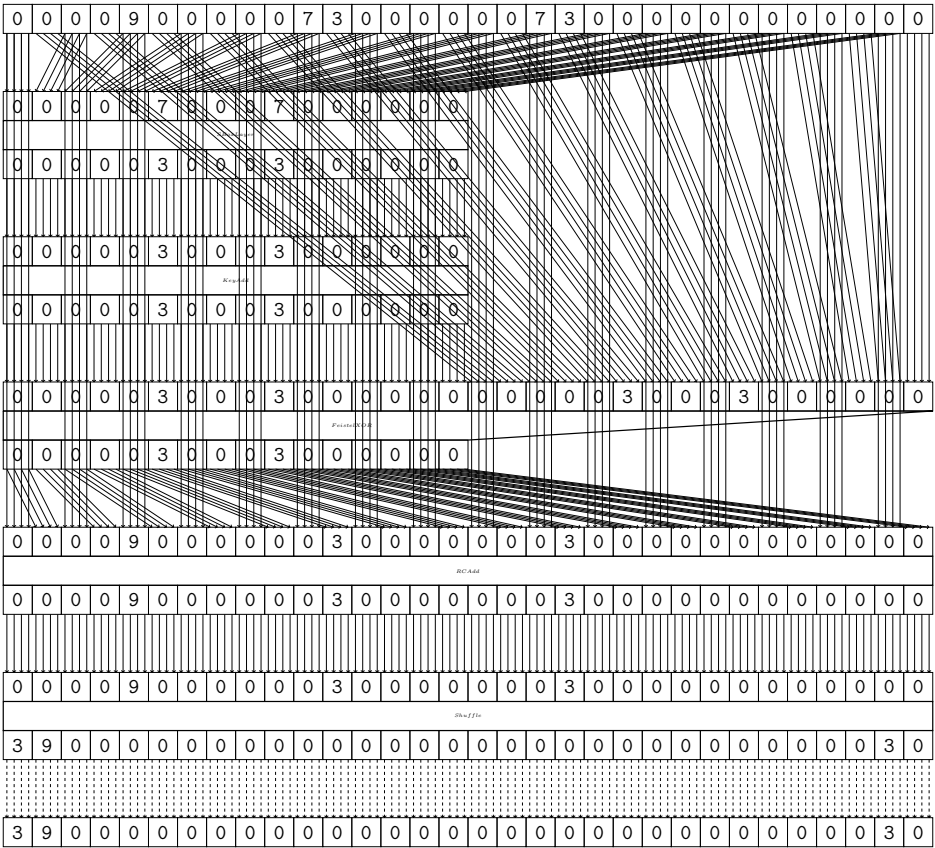
5 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



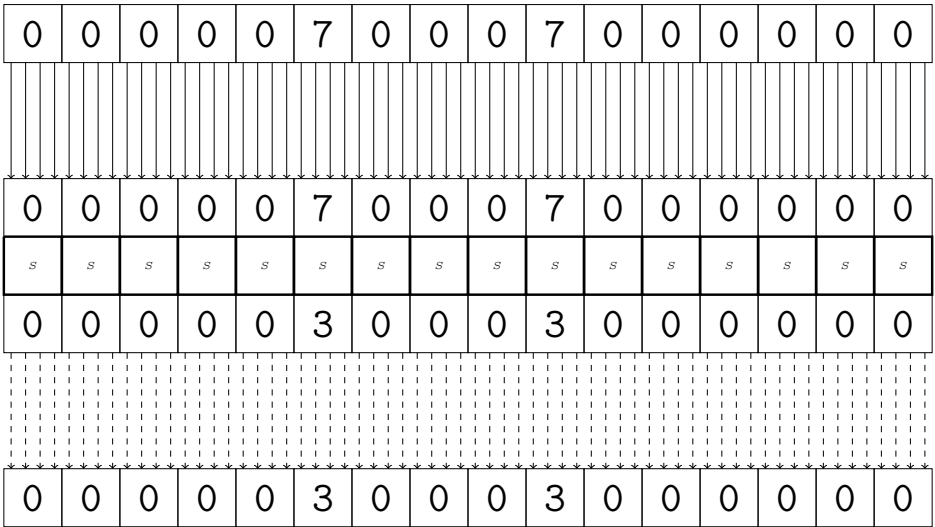
6 warp_round

Maximal linear correlation: $2^{-2.0}$



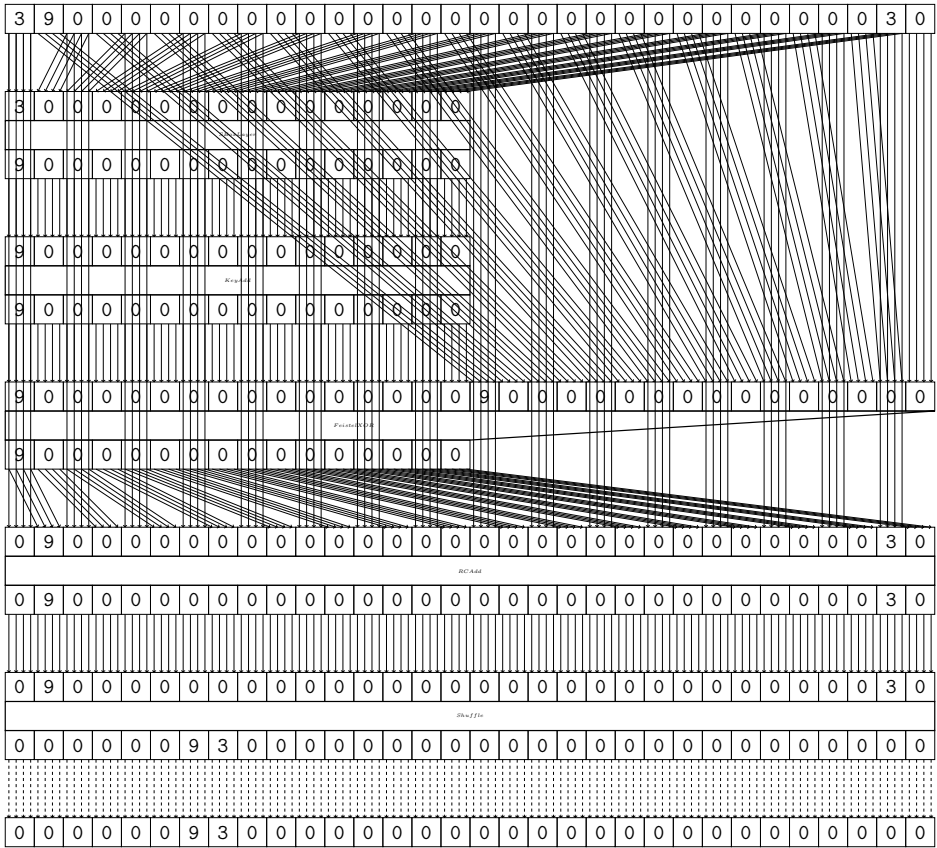
7 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



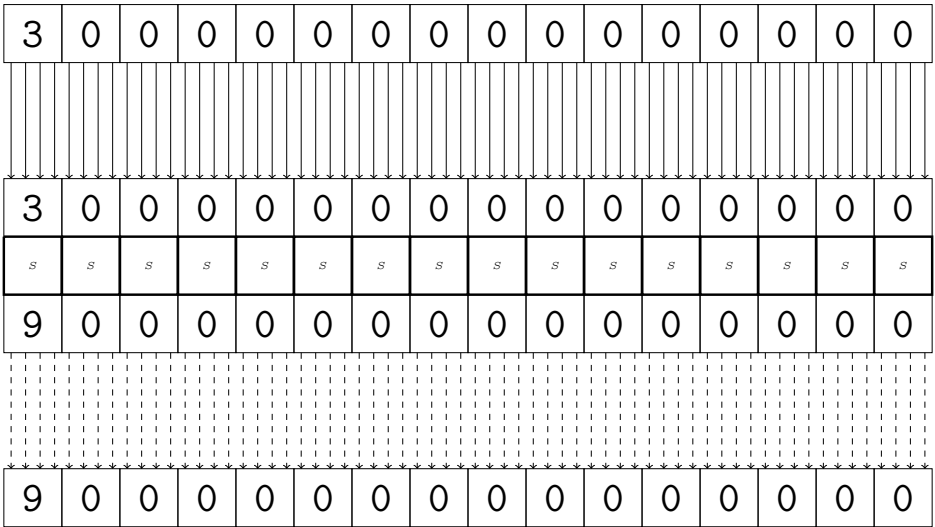
8 warp_round

Maximal linear correlation: $2^{-1.0}$

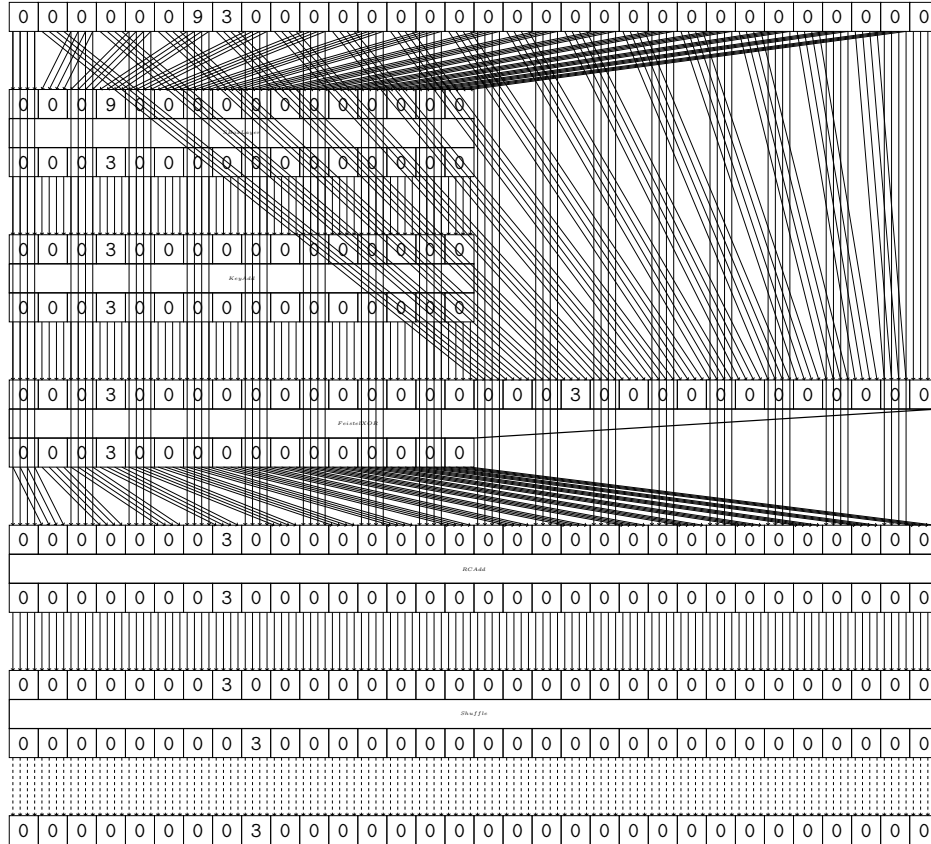


9 SBoxLayer

Maximal linear correlation: $2^{-1.0}$

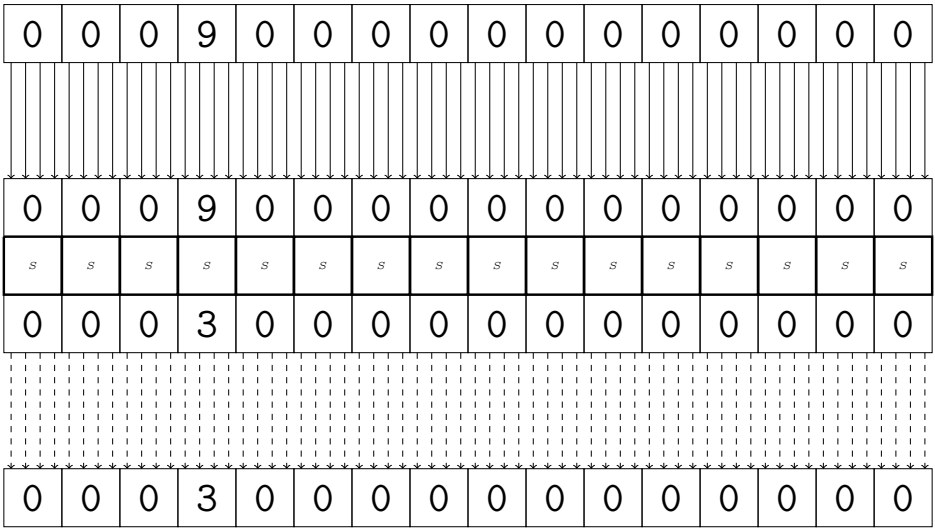


10 warp_round

Maximal linear correlation: $2^{-1.0}$ 

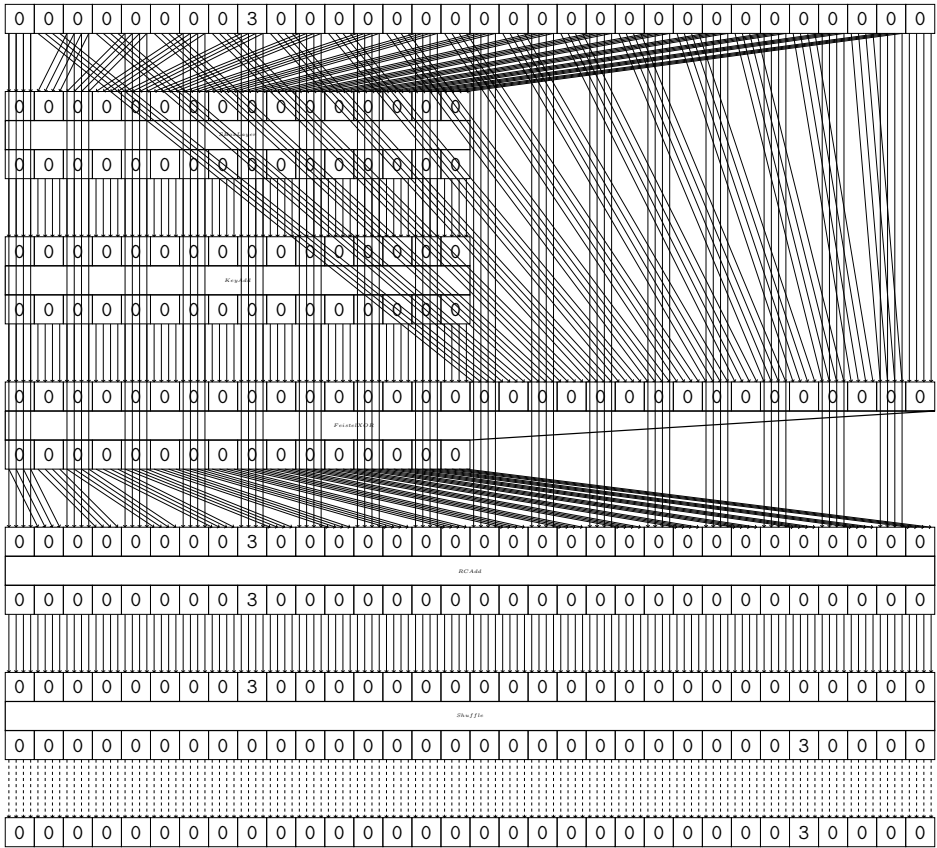
11 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



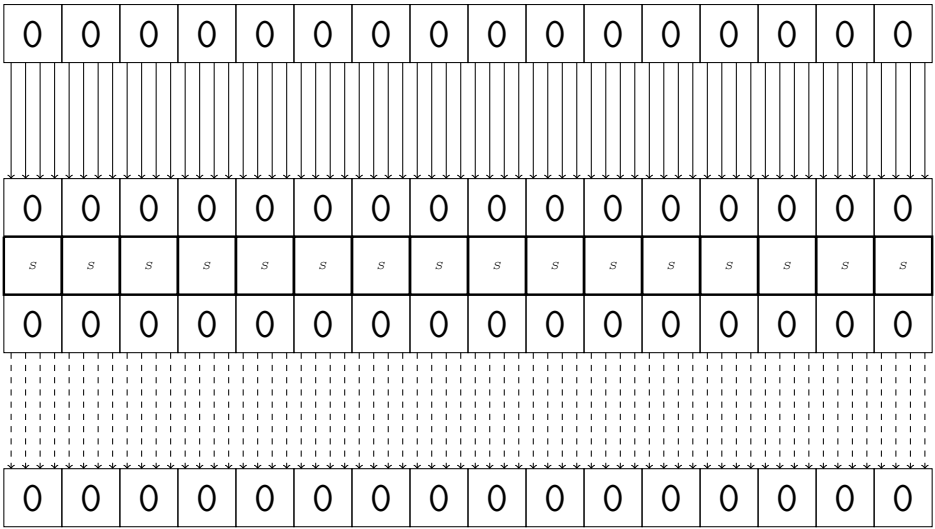
12 warp_round

Maximal linear correlation: $2^{-0.0}$



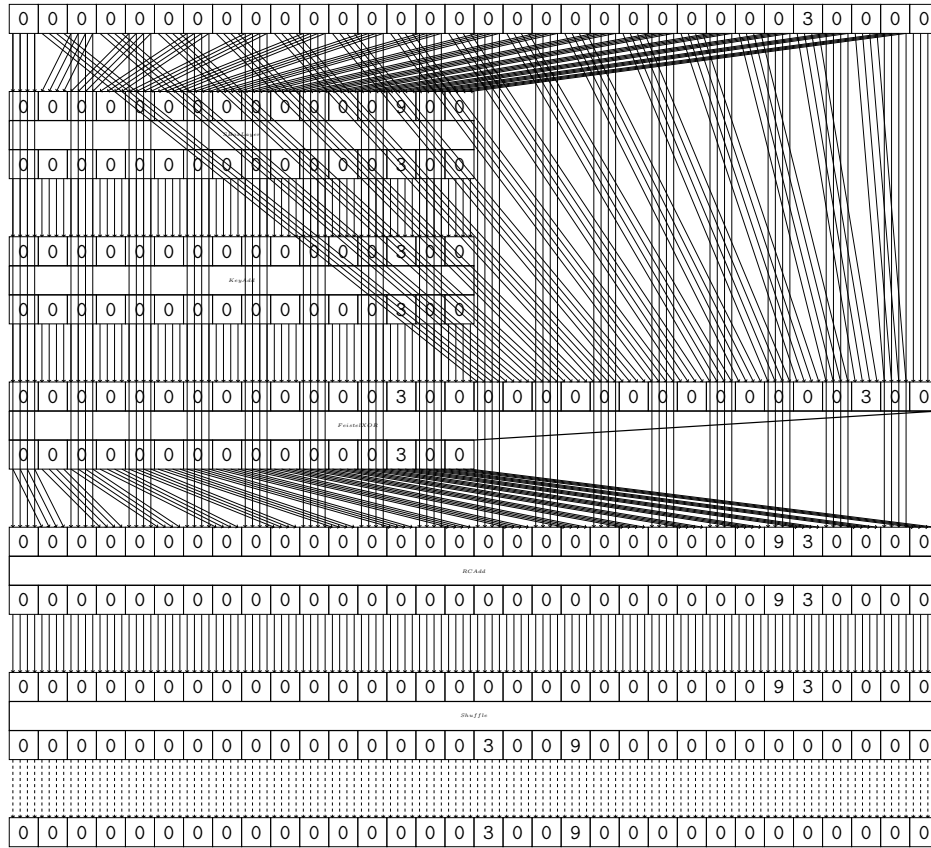
13 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



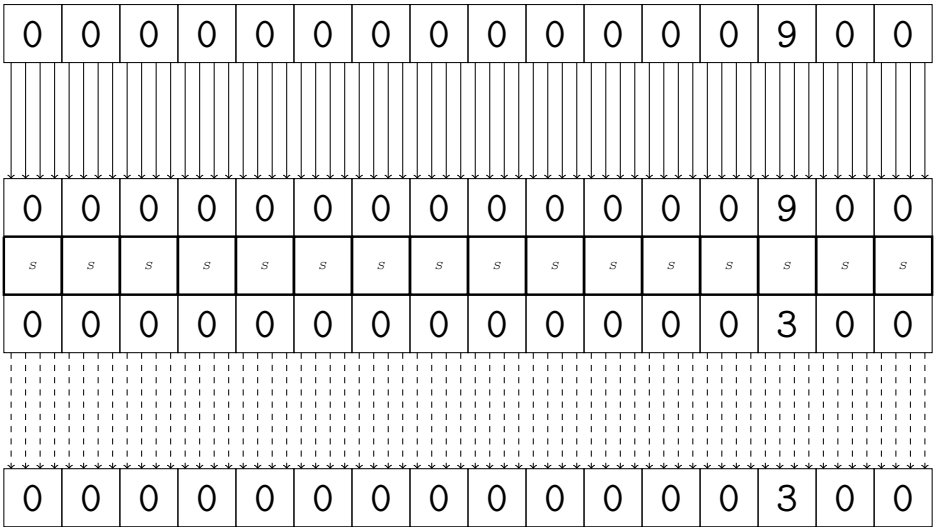
14 warp_round

Maximal linear correlation: $2^{-1.0}$



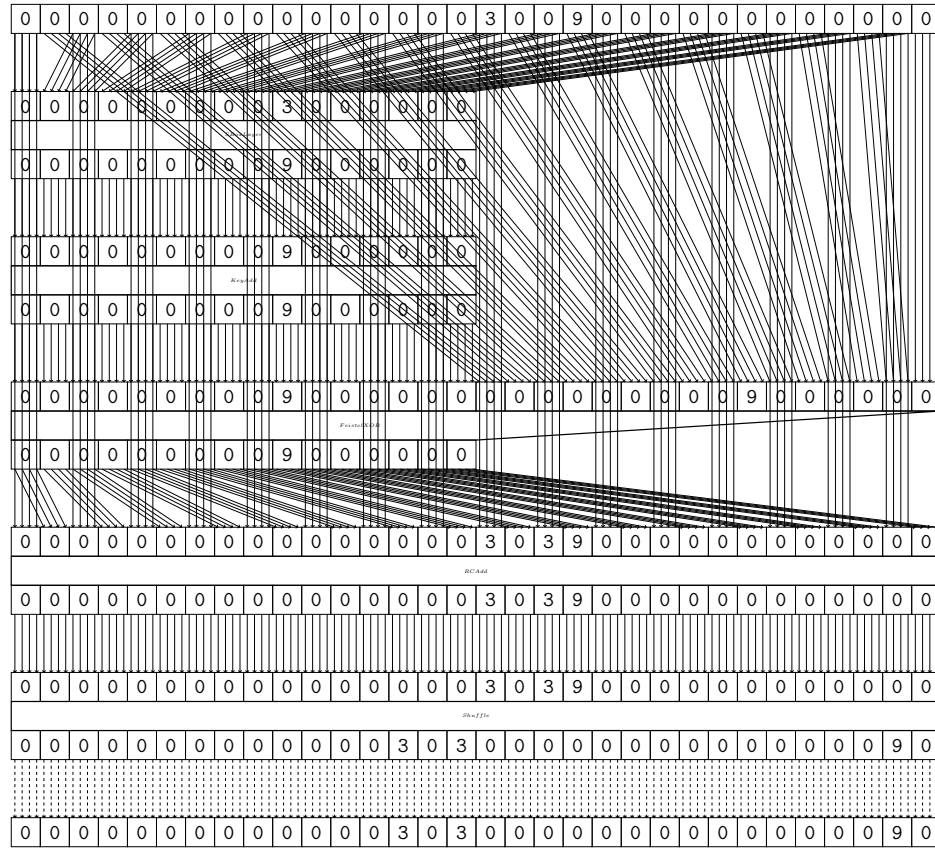
15 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



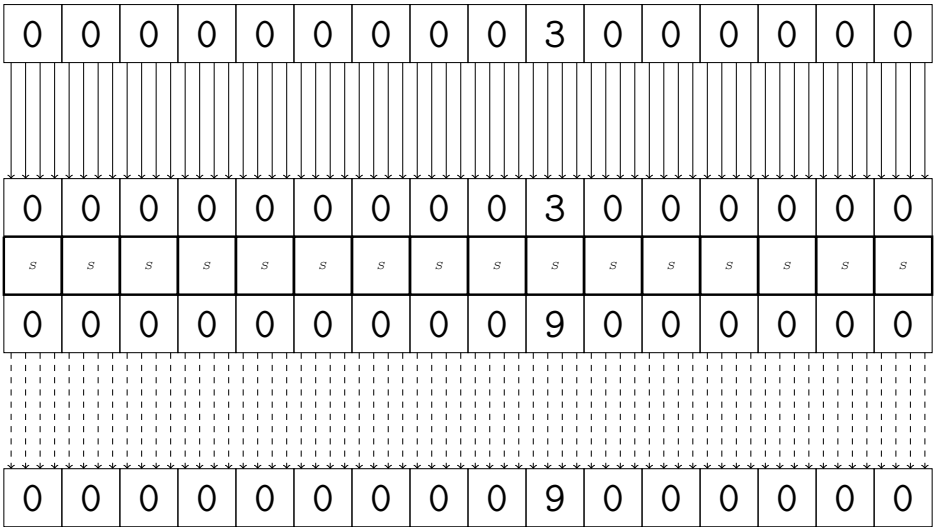
16 warp_round

Maximal linear correlation: $2^{-1.0}$



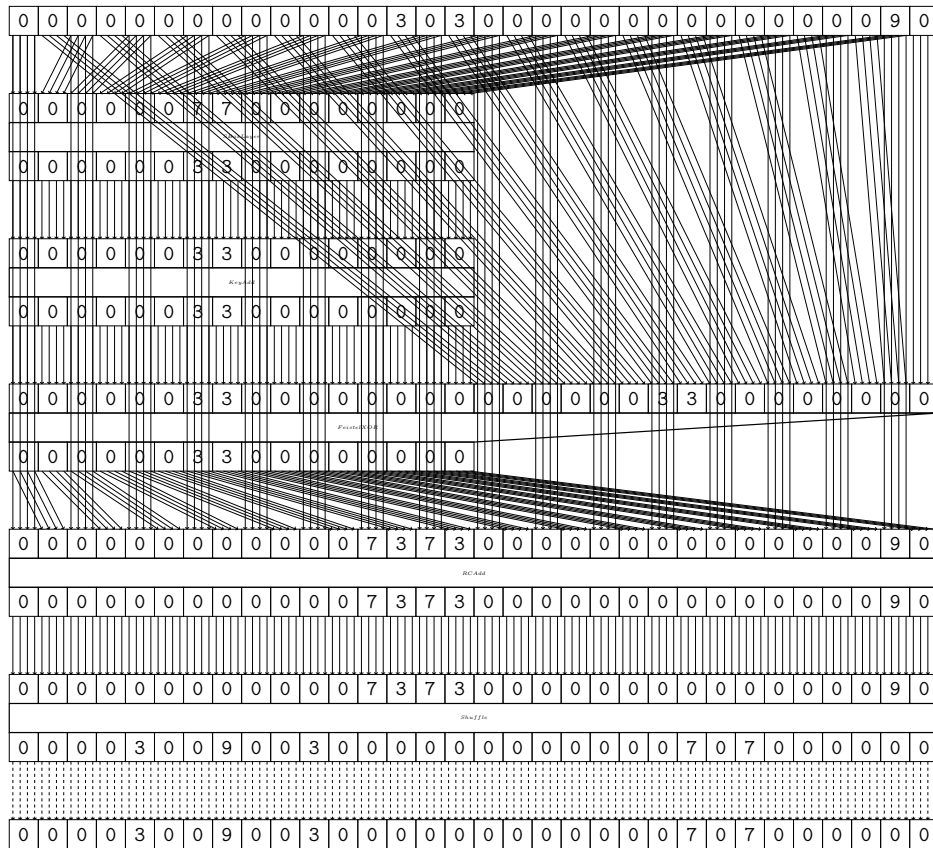
17 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



18 warp_round

Maximal linear correlation: $2^{-2.0}$



19 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

